

Detecting-Persuasion-Techniques-in-Memes

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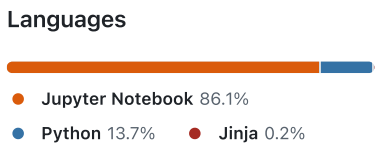
About Code

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Hierarchical, multilingual, multimodal detection of persuasion techniques in memes (SemEval-2024 Task 4), with a custom hierarchical loss and strong H-F1 across 4 languages.

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Hierarchical Detection of Persuasion Techniques in Memes

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Configure

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Configure

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Project

Memes have evolved from entertainment to powerful instruments for online disinformation campaigns. This project aims to build robust models to detect persuasion techniques embedded within memes, enhancing the ability to mitigate the impact of such content on public opinion.

"Detecting Persuasion Techniques in Memes" is a research project aimed at identifying and analyzing various persuasion techniques used in memes. This project is part of the SemEval2024 shared task on "Multilingual Detection of Persuasion Techniques in Memes." The challenge involves understanding and classifying the persuasive elements in memes, which may include logical fallacies and emotional appeals, among others.

Memes are a potent medium in online discourse, often used in disinformation campaigns. By combining text and images, they can effectively influence public opinion. This project seeks to detect these techniques through hierarchical multilabel classification tasks, analyzing both the textual and visual components of memes.

The tasks are divided into:

- **Subtask 1:** Identification of persuasion techniques from the textual content alone.
- **Subtask 2a:** Multimodal analysis, identifying persuasion techniques from both text and image.

This repository contains the code, models developed for the SemEval2024 competition, providing tools to tackle the challenges of detecting persuasion in multimodal content.

Data

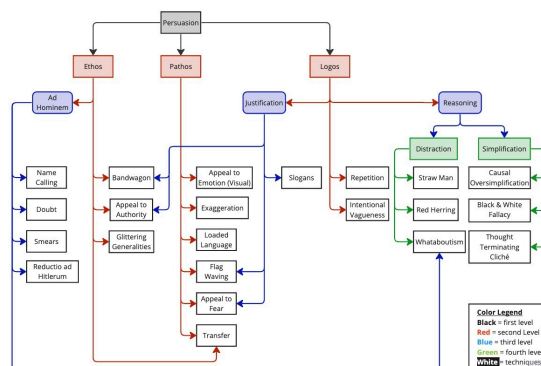
To access the data for this project and submit your predictions, register at the following link:

[SemEval2024 Registration](#)

After registration, you will be able to download the data and participate in the tasks by submitting your predictions through the provided platform.

Techniques

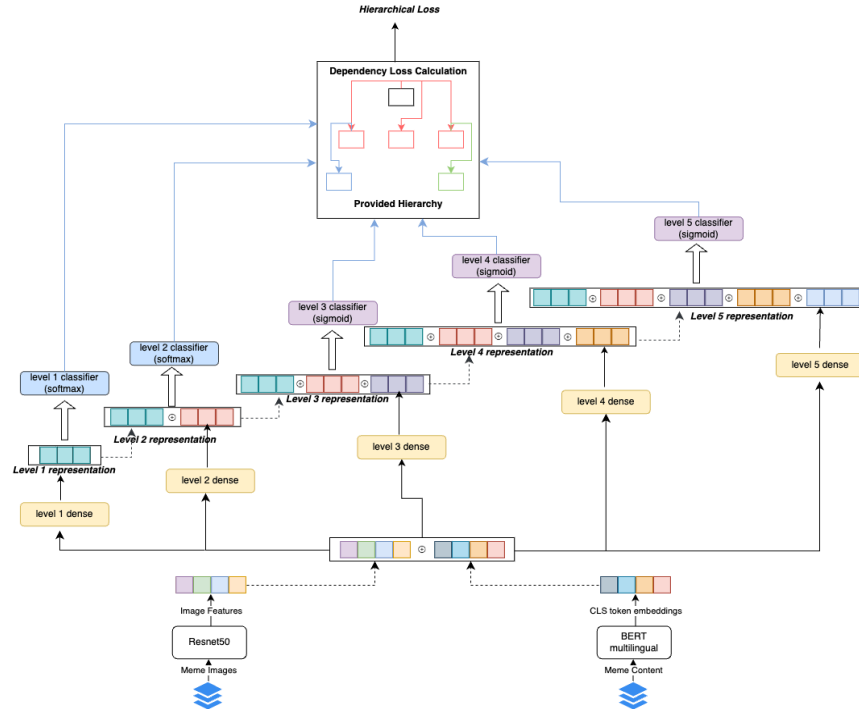
Recent research has identified 22 persuasion techniques applied across textual and visual mediums. This project tackles the challenge of identifying these techniques through a **multi-label hierarchical classification** approach. Given the evolving nature of internet memes and their heavy reliance on current trends and sarcasm, detecting such techniques requires a sophisticated methodology.



Related Work

- **Propaganda Detection:** Traditional binary classifiers like SVM, Naive Bayes, and Random Forest show limitations in identifying complex hate speech.
- **Transformer Models:** Advanced methods such as DeBERTa, XLM-RoBERTa, and GPT-3 have demonstrated improved performance in multimodal setups.
- **Shared Tasks:** Models from competitions like SemEval highlight the importance of transformer-based and multimodal approaches for detecting propaganda.

Architecture



Hierarchical Loss Function

The **hierarchical loss** is designed to maintain consistency across classification levels:

1. **Layer-Specific Loss:** Layer specific loss is straight forward it is summation of losses at each level for predictions made by considering both current level and previous level (parent level) representations. Formulation of this loss at each level is given as:

$$L_{level} = \begin{cases} -\sum_{i=1}^N y_i \log(\hat{y}_i) & \text{for softmax} \\ -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] & \text{for sigmoid} \end{cases}$$

The sum of all layers is defined as:

$$L_{layer\ specific} = \sum_{l=1}^5 L_l$$

2. **Dependency Loss:** Penalizes incorrect predictions that deviate from valid parent-child relationships.

$$D_l = \text{softplus} \left(\sum_{k \in C_{labels}} \delta(k, P_{labels}) \right)$$

where C_{labels} is the set of predicted labels at $level_l$ for instance i , $P_{\{labels\}}$ is the set of predicted labels at $level_{l-1}$ (parent level), and $\delta(k, P_{labels})$ is an indicator function that returns 1 if label k is not a valid child of any labels in $P_l(i)$, and 0 otherwise.

The total dependency loss is then:

$$L_{dependency} = \sum_{l=2}^L ploss_{l-1}^{D_l + l_{prev}} \cdot ploss_l^{D_l + l_{curr}}$$

where $ploss_{l-1}$ and $ploss_l$ are predefined values that escalate with the hierarchy level to impose heavier penalties for misclassifications at higher levels, and β is a scaling factor for the dependency loss's overall impact. Where l_{prev} and l_{curr} are the number of classes that are correctly classified in previous and current level.

Overall, Hierarchical loss is given by

$$H_{loss} = \alpha \cdot L_{layer_specific} + \beta \cdot L_{dependency}$$

where α and β are hyperparameters.

Results

The following table provides the success of Hierarchical loss over conventional BCE loss for multi label classification.

Feature Extractor	Loss Function	Hierarchical F1	Hierarchical Precision	Hierarchical Recall
mBERT	Traditional BCE	0.518	0.615	0.448
mBERT	Hierarchical Loss	0.573  (10.62%)	0.586	0.560  (25%)
OpenAI text-embedding-3-large	Traditional BCE	0.585	0.659	0.526
OpenAI text-embedding-3-large	Hierarchical Loss	0.645  (10.26%)	0.656	0.635  (20.72%)
SOTA	-	0.752 	0.684	0.836

Notes:  indicates improvement over baseline |  indicates best performance | Percentages show relative improvement

Achieved *state-of-the-art results* in Bulgarian and Macedonian languages for SemEval 2024 Task 4 (Subtask 2a).

Language	H-F1 (Subtask 1)	H-F1 (Subtask 2a)
English	0.66391	0.69666
Bulgarian	0.48411	0.65638
Macedonian	0.46615	0.69844
Arabic	0.44478	0.53378

Discussion and Error Analysis

- Misclassification Issues:** Instances where the model predicted incorrect techniques (e.g., predicting “Loaded Language” instead of “Doubt”).
- Multimodal Challenges:** Handling multiple speakers within memes remains a challenge. Further improvements could be made through *data cleaning* and *prompt engineering* with large language models.

Conclusion

This project successfully demonstrates the potential of hierarchical classification for detecting persuasion techniques in memes. With improvements in *loss functions* and *advanced embeddings*, the model achieves state-of-the-art performance. Future work will explore more *robust multimodal integration* and *language expansion* to further enhance meme analysis.

Usage

To reproduce the results of this project, follow these steps to set up the environment and run the implementation.

1. First, clone the project repository from GitHub:

```
git clone https://github.com/iqbal-sk/Detecting-Persuasion-Techniques-in-Memes.git
cd Detecting-Persuasion-Techniques-in-Memes
```

2. It is recommended to use a virtual environment to avoid dependency conflicts. You can create a virtual environment using `venv` :

```
python -m venv env
source env/bin/activate
```

3. Once the virtual environment is activated, install the project dependencies listed in the requirements.txt file:

```
pip install -r requirements.txt
```

4. Configure the Task and Models

Open the configuration file (`config.j2`) and set the task variable and the corresponding models as follows:

- **task:** Choose either `subtask1` or `subtask2a` .
 - For **subtask1**, configure only the `text_model` :
 - Possible values: `"mBERT"` , `"XLNet"` , `"XLMRoBERTa"` , `"OpenAiSmall"` , `"OpenAiLarge"`
 - For **subtask2a**, configure both `text_model` and `image_model` :
 - `text_model` : `"OpenAiSmall"` , `"OpenAiLarge"` , `"mBERT"`
 - `image_model` : `"ResNet50"` , `"CLIP"`

You can adjust the `hyperparameters` to better suit your experiments. Modify the hyperparameters section in the configuration file (`config.j2`)

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